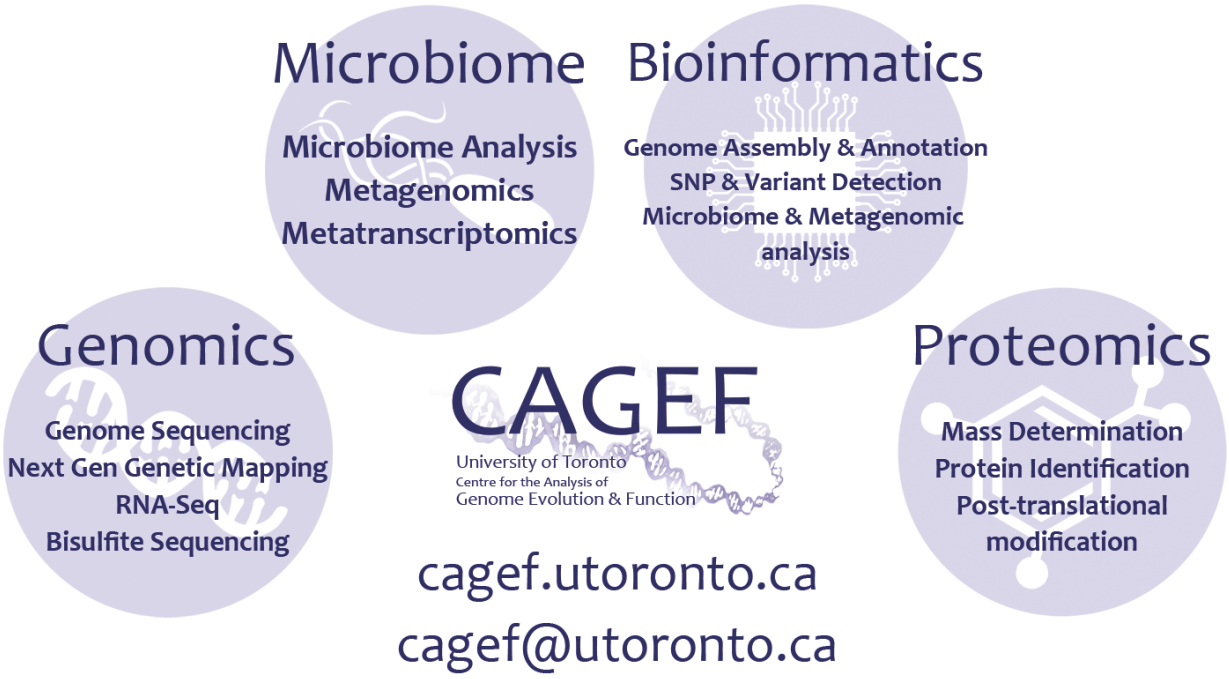



# CSB1021HF LEC0131

# FUNDAMENTALS OF GENOMIC DATA SCIENCE

## 0.0.0 Module 3: Reference alignment and RNA-Seq on Galaxy

## 0.1.0 About Fundamentals of Genomic Data Science

Fundamentals of Genomic Data Science is brought to you by the **Centre for the Analysis of Genome Evolution & Function (CAGEF)** bioinformatics training initiative. This course was developed based on feedback on the needs and interests of the Department of Cell & Systems Biology and the Department of Ecology and Evolutionary Biology.

The structure of this course is a "code-along", hands-on style! A few hours prior to each lecture, materials will be made available for download at QUERCUS (https://q.utoronto.ca/). The teaching materials will consist of a weekly PDF that you can use to follow along with the instructor along with any datasets that you'll need to complete the module. This learning approach will allow you to spend the time coding and not taking notes!

As we go along, there will be some in-class challenge questions for you to solve. Post lecture assessments will also be available for each module, building upon the concepts learned in class (see syllabus for grading scheme and percentages of the final mark).

### 0.1.1 Where is this course going?

We'll take a blank slate approach here to learning genomic data science and assume you know nothing about programming or working directly with next generation sequencing data. From the beginning of this course to the end we want to guide you from potential scenarios like:

- You don't know what to do with a set of raw sequencing files fresh from a facility like CAGEF.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

- You've been handed a legacy pipeline to analyse your data or maintain for the lab, but you don't know what it runs or how.
- You plan on generating high-throughput data but there are no bioinformaticians around to help you out.

and get you to the point where you can:

- Recognize the basic tools in sequence analysis.
- Plan and write your own data analysis pipelines.
- Explain your data analysis methods to labmates, supervisors, and other colleagues.

### 0.1.2 How do we get there?

In the first half of this course, we'll focus on how to generate analysis pipelines using the Galaxy platform – a user-friendly graphical interface that provides access to common sequence analysis tools. After we are comfortable with these tools, we'll look at life through the lens of a command-line interface. It is here that we will learn the basics of file manipulation and how to program scripts that can carry out multiple tasks for us. From there we'll revisit tools from the first half and learn skills to make your data analysis life easier.

## 0.2.0 Goals of the module

1. Learn how to perform and analyze reference alignments with Galaxy.
2. Learn how to perform variant calling with Galaxy.
3. Learn how to analyze RNA-Seq datasets with Galaxy.
4. Visualize reference variant calls and RNA-Seq analyses with the Integrated Genome Viewer.

## 0.3.0 Pre-class modules with Coursera

Each week we strongly encourage you to complete the assigned Coursera modules and/or readings ***before*** class. These are meant to provide you with sufficient background material on each week's module so that we can focus on the act of "doing" something with that data rather than spend a lot of time on the origins of it. You'll find a section outlining the next set of Coursera modules and readings at the end of each module.

**0.3.1** Go to www.coursera.org and sign up for an account with your e-mail.

**0.3.2** Search the following courses and enroll to audit:

- Command Line Tools for Genomic Data Science, Johns Hopkins University.

## 0.4.0 Setting up your working directory

We suggest that you create a new directory (folder) for this course directly off your root directory called "**FGDS**". Working from your root directory is not necessary, but it will make some of the aspects of the course a little easier to manage. For MacOS users, we suggest you create this as a subfolder in your **user** directory.

0.4.1 Within this directory, create another directory called "**Module3**". This is where we will store the data used in this week's module.

0.4.2 Create a subdirectory called "**downloads**" to store the initial files as we download them before decompressing and working with them in later steps.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 1.0.0 Reference alignments in Galaxy

### 1.0.1 Log into usegalaxy.ca

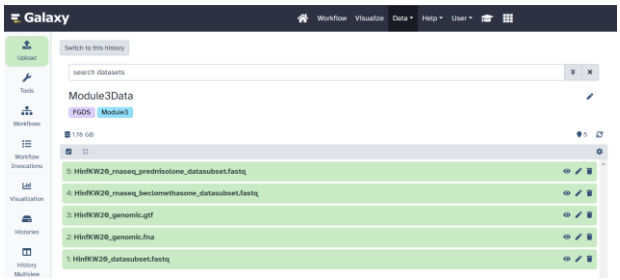
Recall from last week that we started working with this Canadian-based Galaxy server. You can use your University of Toronto login credentials here at https://usegalaxy.ca

## 1.1.0 Add fastq files from shared history

Normally you would upload the files you're interested in working on yourself but for today's reference alignment we are interested in working with some shared files on the server:

- **HinfKW20_datasubset.fastq**: our raw fastq read data subset from last week's lecture
- **HinfKW20_genomic.fna**: a genomic reference nucleotide fasta file for *H. influenza*

1.1.1 *View* the shared data from the *menu > Data > Histories > Shared with me >* [redacted] This shared history will contain the data from the above two data sets along with some additional data that we'll use in today's lecture.

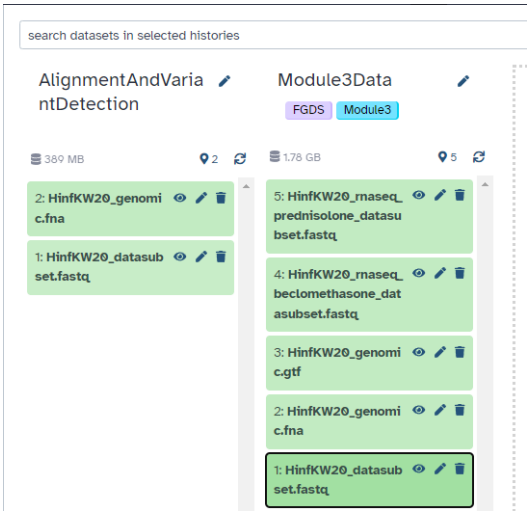



1.1.2 Import the history with the *Import this history* button on the upper-left side of the interface. Rename this imported history [redacted] Leave **Copy only the active, non-deleted datasets** selected.

1.1.3 Go to *History pane > Create* new history and name it **AlignmentAndVariantDetection**.

1.1.4 Go to *Activity pane > History Multiview* and add the [redacted] history using the *Select histories* button.

1.1.5 One at a time, select the two datasets outlined in **1.1.0** from the [redacted] history and drag them to **AlignmentAndVariantDetection**.






University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

1.1.6 Verify that the datatype for **HinfKW20_genomic.fna** is fasta using the *Edit attributes* icon (pencil) for the dataset and checking the *Datatypes* tab.

**Upload your own data**: Recall that you could also upload your own version of the **HinfKW20_genomic.fna** dataset from your Module1 data folder using the *tools pane > Get Data > Upload File from your computer*. You could also retrieve the **HinfKW20_datasubset.fastq** file from your **AssemblyandAnnotation** history created in Module2!

## 1.2.0 Aligning reads to a reference genome

Now that we have our datasets prepared, we want to go about the practice of aligning our raw fastq reads to the reference genome. Recall from last week that the *FastQC* results of our raw data were decent. Although we took the time to use *Trimmomatic* on our dataset, to save on time, we will use the original subset directly for our alignment. For your own workflows, however, you may wish to add a quality filtering step as well.

To map reads to a reference genome, there are a few tools available to us including the **Burrows-Wheeler Aligner** (BWA, circa 2010) and **Bowtie2** (Langmead and Salzberg, Nat Methods 2012) which is part of the "Tuxedo suite" of tools used in RNAseq analysis. Today, for our purposes, we'll be working with Bowtie2 which has better overall metrics than BWA.

1.2.1 Go to the

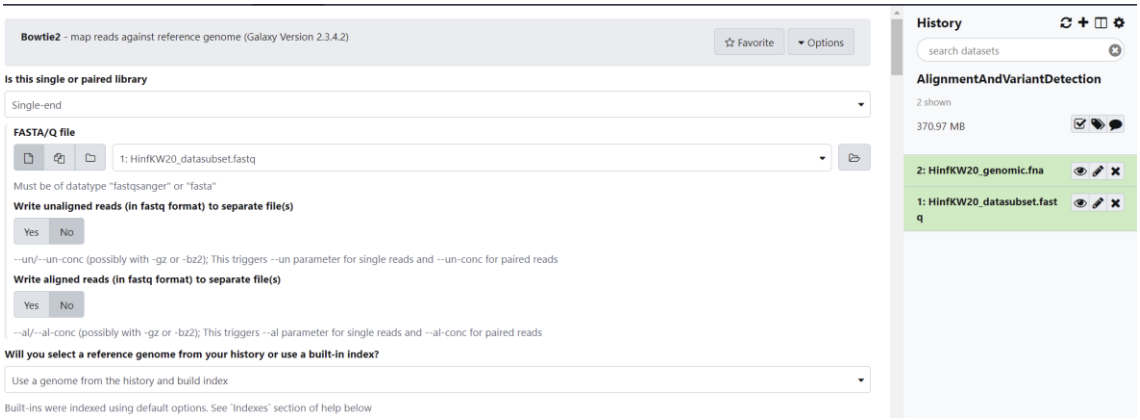
1.2.2 Read the overview for *Bowtie2*. You can revisit the remainder of the *Bowtie2* options later.

1.2.3 Review some of the settings that you can tinker with by changing *Select Analysis Mode* to **Full Parameter List**.

To alter the input options, you will have to switch the dropdown *Do you want to tweak input options?* to **Yes**. Most options should be kept as default unless you are an advanced user and have reason to believe modifying an option would improve your alignment. Returning to the **Default setting only** option there are different pre-sets available for beginners. Proceed with "**No just use defaults**".

1.2.4 Execute a default end-to-end alignment in Bowtie2 after adjusting the following settings:

- Is this single or paired library: **Single-end**
- FASTA/Q file: **HinfKW20_datasubset.fastq**
- Will you select a reference genome from your history or use a built-in index? **Use a genome from the history and build index**
    - Select reference genome:



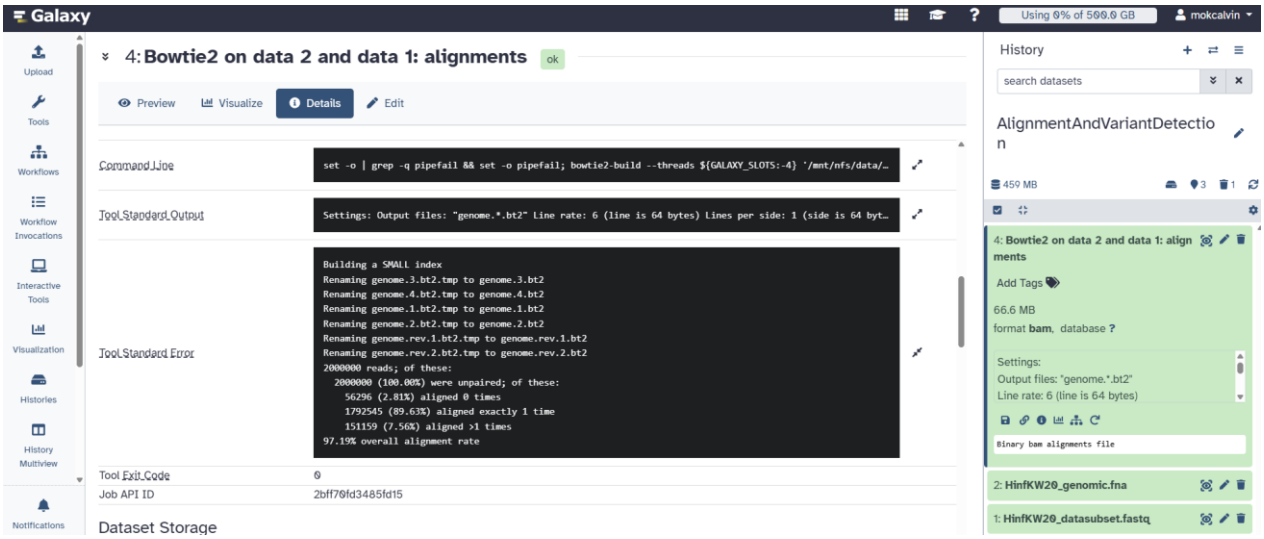



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 1.3.0 Review the alignment stats

Time to take a closer look at the output results and get a sense of how well the reads were aligned to the reference genome.

1.3.1




Here we can retrieve a summary of our alignment run including the overall alignment rate.

## 1.4.0 Review the alignment output

1.4.1 Return to the *History pane > Bowtie 2 on dataset 1 and 2: alignments > View* icon.

1.4.2 At the resulting page, choose the *Raw* tab from the set of choices.

**Caution!** Notice that the entry says the file format is ***BAM*** but we are able to view it in a human readable ***SAM*** format when on Galaxy. If you were to download the data, however, it would not be converted!

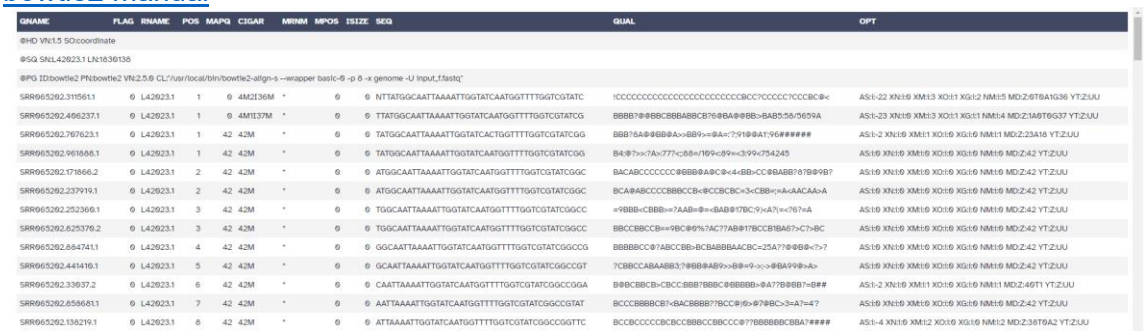
1.4.3 In SAM format, we can see that each read is represented by one line with the columns in these files representing:

- QNAME: Read name
- FLAG: Bitwise flag that categorizes the read (eg. Unmapped or mapped)
- RNAME: Reference sequence (ie your genome name)
- POS: Leftmost mapping position
- MAPQ: Mapping quality
- CIGAR: Cigar string providing information on gaps in the alignment
- MRNM/RNEXT: Reference sequence of the mate (paired-end only)
- MPOS/PNEXT: Reference position of the mate (paired-end only)
- ISIZE/TLEN: Template length based on position of the mate (paired-end only)
- SEQ: Read sequence string
- QUAL: Read quality string (Phred scores)



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

- OPT: Optional field holding meta-information about the aligned read. For more information see the bowtie2 manual



| QNAME | FLAG | RNAME | POS | MAPQ | CIGAR | MRNM | MPOS | ISIZE | SEQ | QUAL | OPT |
|---|---|---|---|---|---|---|---|---|---|---|---|
| @HD VN:1.5 SO:coordinate | | | | | | | | | | | |
| @SQ SN:L42623.1 LN:1636136 | | | | | | | | | | | |
| @PG ID:bowtie2 PN:bowtie2 VN:2.5.6 CL:"/usr/local/bin/bowtie2-align-s --wrapper basic-0 -p 8 -x genome -U input_f.fastq" | | | | | | | | | | | |
| SRR6526239115611 | 0 | L42623.1 | 1 | 0 | 4M2I36M | * | 0 | 0 | NTTATGGCAATTAAAATTGGTATCAATGGTTTTGGTCGTATC | !CCCCCCCCCCCCCCCCCCCCCCCCBCC?CCCCC?CCCBC@< | AS:i:-22 XN:i:0 XM:i:3 XO:i:1 XG:i:2 NM:i:5 MD:Z:6T6A1G36 YT:Z:UU |
| SRR6526240662371 | 0 | L42623.1 | 1 | 0 | 4M1I37M | * | 0 | 0 | TTATGGCAATTAAAATTGGTATCAATGGTTTTGGTCGTATCG | BBBB?@@BBCBBBABBCB?6@BA@@BB>BAB5:56/5659A | AS:i:-23 XN:i:0 XM:i:3 XO:i:1 XG:i:1 NM:i:4 MD:Z:1A9T6G37 YT:Z:UU |
| SRR6526240767231 | 0 | L42623.1 | 1 | 42 | 42M | * | 0 | 0 | TATGGCAATTAAAATTGGTATCACTGGTTTTGGTCGTATCGG | BBB?6A@@BB@A>>BB9>=@A=;?;?1@@A1;96###### | AS:i:-2 XN:i:0 XM:i:1 XO:i:0 XG:i:0 NM:i:1 MD:Z:23A18 YT:Z:UU |
| SRR6526239616861 | 0 | L42623.1 | 1 | 42 | 42M | * | 0 | 0 | TATGGCAATTAAAATTGGTATCAATGGTTTTGGTCGTATCGG | B4;@?>>;?A>777<;68=/169><89=<3;99<754245 | AS:i:0 XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:42 YT:Z:UU |
| SRR6526217181662 | 0 | L42623.1 | 2 | 42 | 42M | * | 0 | 0 | ATGGCAATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGC | BACABCCCCCCC@BBB@A@C@<4<BB>CC@BABB?8?B@9B? | AS:i:0 XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:42 YT:Z:UU |
| SRR6526223791911 | 0 | L42623.1 | 2 | 42 | 42M | * | 0 | 0 | ATGGCAATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGC | BCA@ABCCCCBBBCCB<@CCBCBC=3<CBB=<A<AACAA>A | AS:i:0 XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:42 YT:Z:UU |
| SRR6526225236611 | 0 | L42623.1 | 3 | 42 | 42M | * | 0 | 0 | TGGCAATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGCC | =9BB8<CBBb==?AAB=@=<BAB@17BC;9)<A?(=<?6?=A | AS:i:0 XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:42 YT:Z:UU |
| SRR6526226253762.2 | 0 | L42623.1 | 3 | 42 | 42M | * | 0 | 0 | TGGCAATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGCC | BBCCBBCCB==9BC@6%?AC??AB@1?BCCB1BA6?>C?>BC | AS:i:0 XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:42 YT:Z:UU |
| SRR6526228647411 | 0 | L42623.1 | 4 | 42 | 42M | * | 0 | 0 | GGCAATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGCCG | BBBBBCC@?ABCCBB>BCBABBBAACBC=25A??@@B@<?>? | AS:i:0 XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:42 YT:Z:UU |
| SRR6526224414161 | 0 | L42623.1 | 5 | 42 | 42M | * | 0 | 0 | GCAATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGCCGT | ?CBBCCABAABB3;?9BB@AB9>>B@=9-><>@BA99@>A> | AS:i:0 XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:42 YT:Z:UU |
| SRR6526233393722 | 0 | L42623.1 | 6 | 42 | 42M | * | 0 | 0 | CAATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGCCGGA | B@BCBBCB>CBCCBBB?BBBC@BBBBB>@A??B@BB?=B## | AS:i:-2 XN:i:0 XM:i:1 XO:i:0 XG:i:0 NM:i:1 MD:Z:46T1 YT:Z:UU |
| SRR6526225866811 | 0 | L42623.1 | 7 | 42 | 42M | * | 0 | 0 | AATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGCCGTAT | BCCCBBBBCB?<BACBBBB??BCC@)6>@?@BC>3=A?=4? | AS:i:0 XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:42 YT:Z:UU |
| SRR6526213821911 | 0 | L42623.1 | 8 | 42 | 42M | * | 0 | 0 | ATTAAAATTGGTATCAATGGTTTTGGTCGTATCGGCCGGTTC | BCCBCCCCCCBCBCCBBBCCBBCCC@??BBBBBBCBBA?#### | AS:i:-4 XN:i:0 XM:i:2 XO:i:0 XG:i:0 NM:i:2 MD:Z:36T0A2 YT:Z:UU |

1.4.4 Choose the ***Visualize*** tab from the set of choices. This will take you to a page where you can select a web-based version of IGV which you can use to navigate your reads much like we did in Module 01.

1.4.5 In the top right of this IGV interface, you can **Expand** the options from the blue circle with double-arrows. On the Settings tab of this side menu, you can set the **"Source"** dropdown to **"Galaxy History Dataset"** and then select **"HinfKW20_genomic.fna"**. This will allow you to view and zoom in on the various alignments.

Now that we've generated our first reference alignment, it's time to proceed to variant calling!

**SAM and BAM**: Sequence alignment/map (SAM) and binary alignment/map (BAM) are common file formats in bioinformatics that are produced as the result of a sequence alignment and originated in 2009 from Heng Li and Bob Handsaker et al. It has since expanded to include various features and is the standard for storing mapped data results across various pipelines and platforms.






University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 2.0.0 Variant calling with SAMtools and BCFtools

Now that we've generated our alignment and have a binary version of a sequence alignment map, we can proceed with assessing the alignment for the presence of SNVs, indels, or even inversions. We'll begin by examining the result of an **mpileup**, which was briefly described in Module 1.

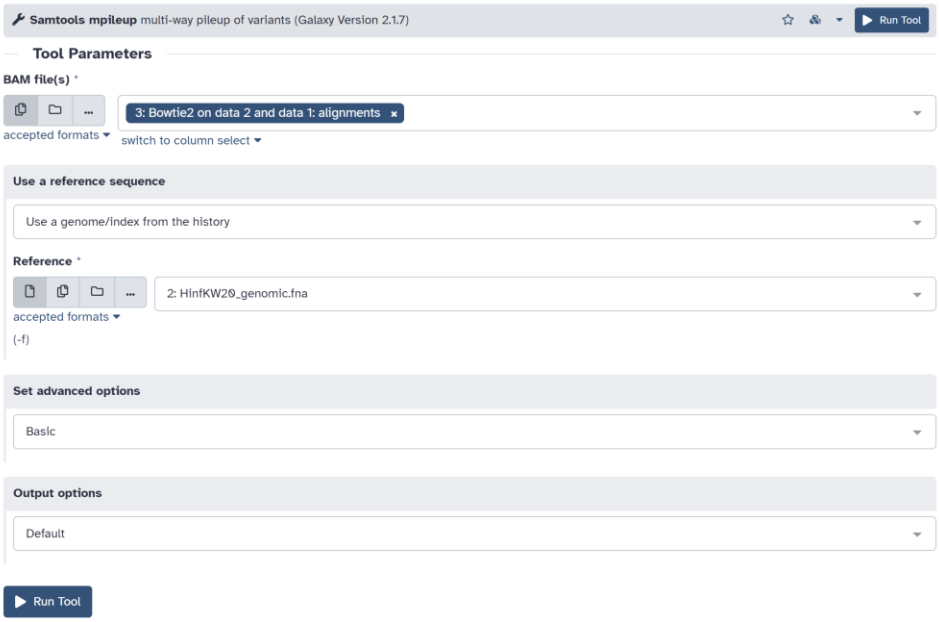
## 2.1.0 Generate an *mpileup* of your alignment

2.1.1 Go to *tools pane* > [redacted]

2.1.2 Scroll down to review the purpose and input/output information for *samtools mpileup*.

2.1.3 Execute *samtools mpileup* with the following settings:

- BAM file(s): **Bowtie2 on dataset 1 and 2: alignments**
- Choose the source for the reference genome: **Use a genome/index from the history**
- Using reference genome: **HinfKW20_genomic.fna**
- Set advanced options: Basic
- Output Options: Default




## 2.2.0 Reviewing *mpileup* results

Time to review the results of the pileup we've generated. It will give us a position-by-position summary of sequence calls based on the aligned reads we generated from *Bowtie2*.

2.2.1 Click on *History pane > samtools mpileup on dataset 2 and 3 > View* icon. The default ***Preview*** tab will bring up a table version of the mpileup on Galaxy. The 6-column format represents the following:

1. Reference chromosome/contig
2. Reference position
3. Reference base
4. Total reads covering the site
5. String of all aligned bases from overlapping fragments



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

| Category | Forward (+) direction fragment | Reverse (-) direction fragment |
|---|---|---|
| Start of a new fragment<br><br>* BAQ = Base alignment quality. Calculate ASCII value – 33 to obtain the score. | ^[ASCII-BAQ]<br>eg ^! | ^[ ASCII-BAQ]<br>eg ^K |
| End of a fragment | $ | $ |
| Correct match with reference base | . | , |
| Mismatch with reference, showing the mismatch base | ACGTN | acgtn |
| Insertion with the inserted sequence. | +N[ACGTN] | +N[acgtn] |
| Deletion with the deleted sequence | -N[ACGTN] | -N[acgtn] |

6. String of base qualities (ie Phred scores) for each fragment at that position

| Column 1 | Column 2 | Column 3 | Column 4 | Column 5 | Column 6 |
|---|---|---|---|---|---|
| L42023.1 | 1 | T | 3 | ^!.^K.^K. | 1BB |
| L42023.1 | 2 | A | 4 | ..^K.^K. | B4BB |
| L42023.1 | 3 | T | 6 | ....^K.^K. | B;AC=B |
| L42023.1 | 4 | G | 7 | .....^K. | ?@CA9BB |
| L42023.1 | 5 | G | 10 | .....^K. | C@8?A@BCB? |
| L42023.1 | 6 | C | 11 | .....^K. | C@A>BABCBCB |
| L42023.1 | 7 | A | 12 | .....^K. | CB@>CBBBBB@B |
| L42023.1 | 8 | A | 13 | .....^K. | CB@:CC<BBBBCB |
| L42023.1 | 9 | T | 14 | .....^K. | CCB?CCCCCCCCCB |
| L42023.1 | 10 | T | 15 | .....^K. | CBBACCBCCCBCCBA |
| L42023.1 | 11 | A | 16 | .....^K. | CB@>CCBB@ABBBCB? |
| L42023.1 | 12 | A | 17 | .....^K. | CBA:CBB=?BCBCBBA; |
| L42023.1 | 13 | A | 19 | .....^K.^K. | CA>7CB>=AABBCCCB;== |
| L42023.1 | 14 | A | 22 | .....^K.^K.^K. | CB>7@B=9BA>BCCCB=>>BBB |
| L42023.1 | 15 | T | 23 | .....^K. | CBB?BC?BCBCCCCCBBCBCCCB |
| L42023.1 | 16 | T | 24 | .....^K. | CCB<BCACCBBBCCCB@BBCCBBB |
| L42023.1 | 17 | G | 25 | .....^K. | CB9;BBA@B3C?B=9BA?BCBBBCB |
| L42023.1 | 18 | G | 25 | ..... | C?>:@<B0B;C<C@9BB>C?BAABA |

## 2.3.0 Convert BAM files to a variant call format

Although the *mpileup* file is a useful way to quickly glance around the genome, it does not necessarily summarize all the variants in a friendly way, which reads support the presence of variants or indels? We can, however, convert our *Bowtie2* BAM file to the ***variant call format*** (VCF) which further summarizes the potential variants across the genome while accounting for specific parameters like genome coverage. These can also be imported by other programs like IGV as we'll see in later steps. To accomplish this task, we'll use a related package to *SAMtools* called *BCFtools*. The purpose of this package is to handle variant calling and the manipulation of variant data.

2.3.1 Click on *tools pane* > [redacted]

2.3.2 Scroll down to review the purpose and input/output information for *bcftools mpileup*.

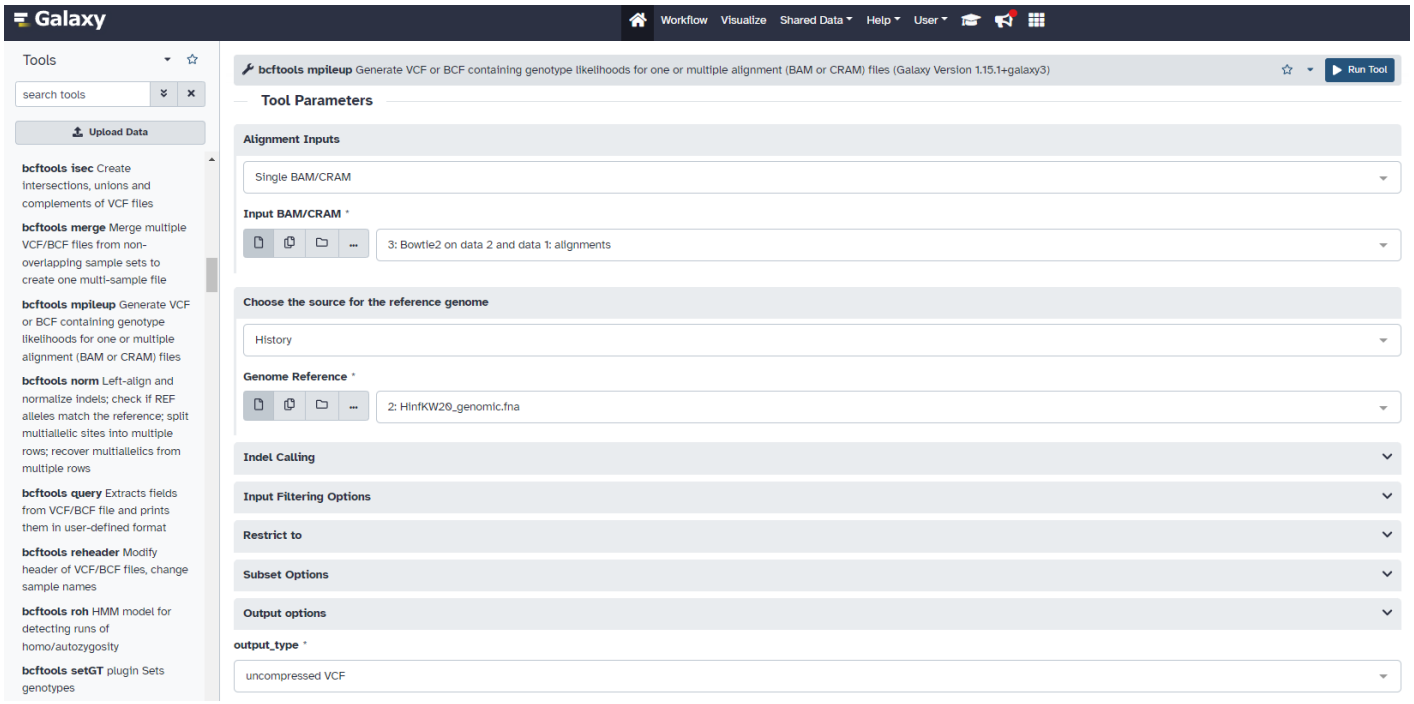
Note that despite its name, this is not creating an *mpileup* for us but rather a **VCF** file. The naming is a legacy to when *SAMtools* used to provide a VCF tool.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

2.3.3 Execute *bcftools mpileup* with the following settings:

- Alignment inputs: **Single BAM/CRAM**
- Input BAM/CRAM: **Bowtie2 on dataset 1 and 2: alignments**
- Choose the source for the reference genome: **History**
    - Genome Reference: **HinfKW20_genomic.fna**
- Output_type:
- All other settings: **default**




## 2.4.0 Review your VCF output

With the variant calling completed, a new VCF file is available in the *History pane*.

2.4.1 Click on *History pane > bcftools mpileup on dataset 2 and 3 > View* icon. This will bring up a view to the VCF file on Galaxy. The start of the file contains metadata represented as a series of lines with **##** at the beginning. Metadata can contain information like how the VCF file was generated (for reproducibility!) or how to interpret column codes. The metadata is followed by a final single **#** line which includes the column names. The 10 columns of the header represent the following:

1. **CHROM**: Reference chromosome/contig
2. **POS**: Reference position. Note that there may be multiple entries with the same position.
3. **ID**: Identifier of the variant. A semicolon-separated list of unique identifiers where available.
4. **REF**: Reference base(s)
5. **ALT**: Alternative base(s) or <*> (homozygous reference site)
6. **QUAL**: A Phred-scaled quality score for the assertion made in “Alt”.
7. **FILTER**: Filter status of the call. Usually PASS if the position has passed all filters.
8. **INFO**: A semicolon-separated list of additional information with codes listed in the metadata.
9. **FORMAT**: An optional genotype information format descriptor consisting of many colon-separated metrics. The codes describe the type of data and its order in the remaining columns.
10. **DATA**: Genotype data specific to metric(s) if specified in **FORMAT**.

* Missing values are specified with a dot (.). See VCF 4.2 specifications section 1.4.2 for more genotype information.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

##INFO=<ID=IMF,Number=1,Type=Float,Description="Maximum fraction of raw reads supporting an indel">

##INFO=<ID=DP,Number=1,Type=Integer,Description="Raw read depth">

##INFO=<ID=VDB,Number=1,Type=Float,Description="Variant Distance Bias for filtering splice-site artefacts in RNA-seq data (bigger is better)",Version="3">

##INFO=<ID=RPBZ,Number=1,Type=Float,Description="Mann-Whitney U-z test of Read Position Bias (closer to 0 is better)">

##INFO=<ID=MQBZ,Number=1,Type=Float,Description="Mann-Whitney U-z test of Mapping Quality Bias (closer to 0 is better)">

##INFO=<ID=BQBZ,Number=1,Type=Float,Description="Mann-Whitney U-z test of Base Quality Bias (closer to 0 is better)">

##INFO=<ID=MQSBZ,Number=1,Type=Float,Description="Mann-Whitney U-z test of Mapping Quality vs Strand Bias (closer to 0 is better)">

##INFO=<ID=SCBZ,Number=1,Type=Float,Description="Mann-Whitney U-z test of Soft-Clip Length Bias (closer to 0 is better)">

##INFO=<ID=FS,Number=1,Type=Float,Description="Phred-scaled p-value using Fisher's exact test to detect

##INFO=<ID=SGB,Number=1,Type=Float,Description="Segregation based metric.">

##INFO=<ID=MQ0F,Number=1,Type=Float,Description="Fraction of MQ0 reads (smaller is better)">

##INFO=<ID=I16,Number=16,Type=Float,Description="Auxiliary tag used for calling, see description of bcf_callret1_t in bam2bcf.h">

##INFO=<ID=QS,Number=R,Type=Float,Description="Auxiliary tag used for calling">

##FORMAT=<ID=PL,Number=G,Type=Integer,Description="List of Phred-scaled genotype likelihoods">

| #CHROM | POS | ID | REF | ALT | QUAL | FILTER | INFO |
|---|---|---|---|---|---|---|---|
| L42023.1 | 1 | . | T | <*> | 0 | . | DP=4;I16=3,0,0,0,82,2434,0,0,84,3528,0,0,0,0,0,0;QS=1,0;FS=0;MQ0F=0.25 |
| L42023.1 | 2 | . | A | <*> | 0 | . | DP=6;I16=4,0,0,0,118,3628,0,0,168,7056,0,0,2,2,0,0;QS=1,0;FS=0;MQ0F=0 |
| L42023.1 | 3 | . | T | <*> | 0 | . | DP=8;I16=6,0,0,0,186,5818,0,0,252,10584,0,0,6,10,0,0;QS=1,0;FS=0;MQ0F=0 |
| L42023.1 | 4 | . | G | <*> | 0 | . | DP=9;I16=7,0,0,0,217,6795,0,0,294,12348,0,0,12,28,0,0;QS=1,0;FS=0;MQ0F=0 |

## 2.5.0 Filter VCF for variants only

You may notice from your output that you are getting base calls at ***every position***. While good for confirmation, again there is an excess of information that we do not need. Instead, we want to filter the variants based on some criteria such as their likelihood, and we only want to view ***non-reference variants*** in the case of output from *bcftools mpileup*.

2.5.1 [redacted]

2.5.2 Scroll down to review the purpose and input/output information for *bcftools call*.

2.5.3 Execute *bcftools call* with the following settings:

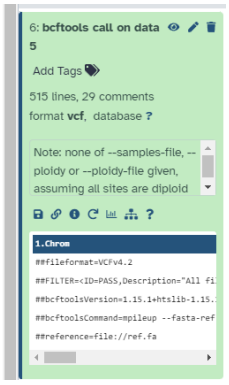
- VCF/BCF Data: **bcftools mpileup on dataset 2 and 3**
- Input/output Options
  - [redacted]
- output_type: **uncompressed VCF**
- All other settings: **default**

2.5.4 Review your output with *History pane > bcftools call on dataset 5 > View.*

Notice now that our data has been filtered to include only variant information.

##bcftools_callCommand=call -m --prior 0.0011 --variants-only --output-type v --threads 1 Input.vcf.gz; Date=Mon Jun 3 16:16:46 2024

| #CHROM | POS | ID | REF | ALT | QUAL | FILTER | INFO |
|---|---|---|---|---|---|---|---|
| L42023.1 | 1056 | . | CG | CGG | 80.2092 | . | INDEL;IDV=5;IMF=0.5;DP=10;VDB=0.604698;SGB=-0.616816;RPBZ=-1.5667;MQBZ=-2.886 |
| L42023.1 | 3808 | . | T | C | 225.417 | . | DP=34;VDB=0.0471191;SGB=-0.693132;MQSBZ=-0.476756;FS=0;MQ0F=0;AC=2;AN=2;DP4 |
| L42023.1 | 20891 | . | AG | AGG | 162.172 | . | INDEL;IDV=39;IMF=0.886364;DP=44;VDB=0.568577;SGB=-0.693144;RPBZ=-0.722435;M |
| L42023.1 | 29688 | . | A | T | 225.417 | . | DP=38;VDB=0.411578;SGB=-0.693143;MQSBZ=0.750424;FS=0;MQ0F=0;AC=2;AN=2;DP4= |
| L42023.1 | 32329 | . | GAAAAAA | GAAAAAAA | 135.341 | . | INDEL;IDV=17;IMF=0.708333;DP=24;VDB=0.237631;SGB=-0.696438;RPBZ=-1.11435;MQB |
| L42023.1 | 36715 | . | G | A | 228.413 | . | DP=43;VDB=0.632997;SGB=-0.693146;RPBZ=-0.887444;MQBZ=2.11693;MQSBZ=-2.12656 |
| L42023.1 | 45496 | . | G | A | 225.417 | . | DP=29;VDB=0.737192;SGB=-0.693079;MQSBZ=-0.670288;FS=0;MQ0F=0;AC=2;AN=2;DP4 |
| L42023.1 | 46361 | . | A | T | 225.417 | . | DP=32;VDB=0.0499261;SGB=-0.69312;MQSBZ=2.29797;FS=0;MQ0F=0;AC=2;AN=2;DP4=0 |
| L42023.1 | 47284 | . | A | T | 225.417 | . | DP=96;VDB=0.377221;SGB=-0.693147;MQSBZ=0.925737;FS=0;MQ0F=0;AC=2;AN=2;DP4= |
| L42023.1 | 52066 | . | CAAAA | CAAAAA | 210.292 | . | INDEL;IDV=56;IMF=0.823529;DP=68;VDB=0.0587162;SGB=-0.693147;RPBZ=-2.7777;MQB |
| L42023.1 | 52337 | . | G | A | 225.417 | . | DP=77;VDB=0.152445;SGB=-0.693147;MQSBZ=-2.9262;FS=0;MQ0F=0;AC=2;AN=2;DP4=0, |
| L42023.1 | 52396 | . | G | A | 225.421 | . | DP=48;VDB=0.444947;SGB=-0.693147;MQSBZ=-2.6082;FS=0;MQ0F=0;AC=2;AN=2;DP4=( |






University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 2.6.0 Download your VCF data

As with our data from Module 2 we can download our VCF file here. Looking in the History pane, you can find details about the file size.

2.6.1 Return to your output list on the *History pane* and select the *Select Items* (checkmark) icon.

2.6.2 Select the following datasets.

- *Bowtie2* on dataset 1 and 2: alignments
- *bcftools call* on dataset 5

2.6.3 Select *History pane > 2 of 6 selected > Auto Build List*.

2.6.4 Name the dataset list **RefAlignDownload** and unselect the *Remove file extensions?* and *Hide original elements* buttons

2.6.5 Use the *Build* button.

2.6.6 Select *History pane > RefAlignDownload > Download*.

2.6.7 Save the .zip file to **~/FGDS/Module3/downloads** (~68 Mb).

2.6.8 Decompress the contents of your .zip file to **~/FGDS/Module3**.

2.6.9 Rename your files using the following table

| Download name | New name |
|---|---|
| Bowtie2 on data 2 and data 1_ alignments.bam | HinfKW20_refalign.bam |
| Bowtie2 on data 2 and data 1_ alignments.bam.bai | HinfKW20_refalign.bam.bai |
| bcftools call on data 5 | HinfKW20_refalign.VCF |

## 2.7.0 Save your history as a workflow

Now that you've completed your first reference alignments and variant calling analysis, you can save your history as a new workflow for future use. You can come back and edit the finer details of the workflow at a later time.

2.7.1 Go to *History pane > History options > Extract Workflow*

2.7.2 Name your workflow **AlignmentAndVariantDetection** and review the tools/input files that are going to be used in the workflow.

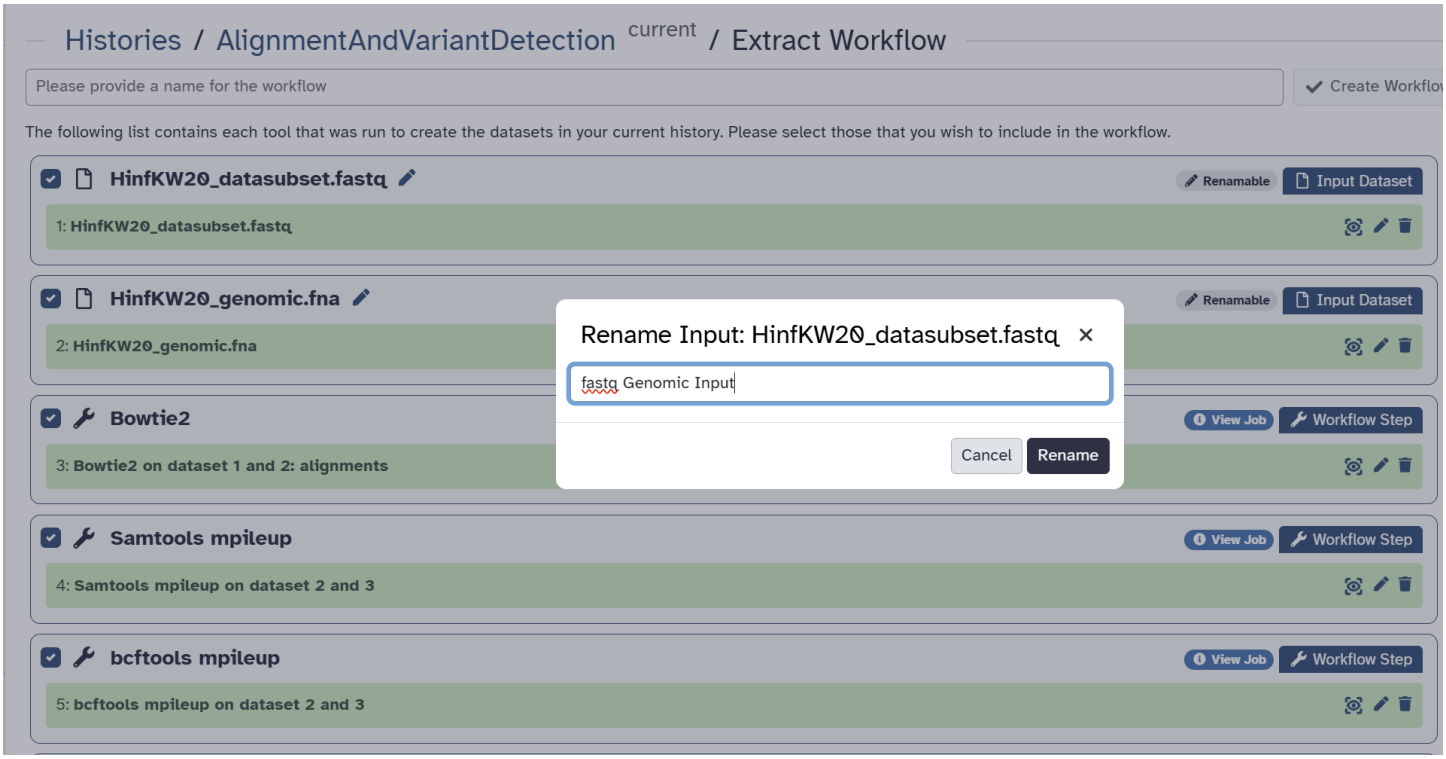
2.7.3 Rename your initial inputs like "HinfKW20_datasubset.fastq" to "fastq WGS input" and "HinfKW20_genomic.fna" to "fna Genomic Reference". These make your input names more descriptive and broad for future uses.

2.7.4 Deselect the [redacted] you've created so that it won't become part of the Workflow.

2.7.5 Once you are ready, select *Create Workflow*.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA






University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 3.0.0Viewing your reference alignment in IGV

Now that we've downloaded our reference alignment and VCF data from Galaxy, we can use a graphical interface like IGV to view the read alignments themselves using the VCF information.

## 3.1.0 Loading your files in IGV

3.1.1 Open up IGV and, if necessary, load your genome again from Module 1 using *Genomes > Load Genome from File >* **HinfKW20_genomic.fna**.

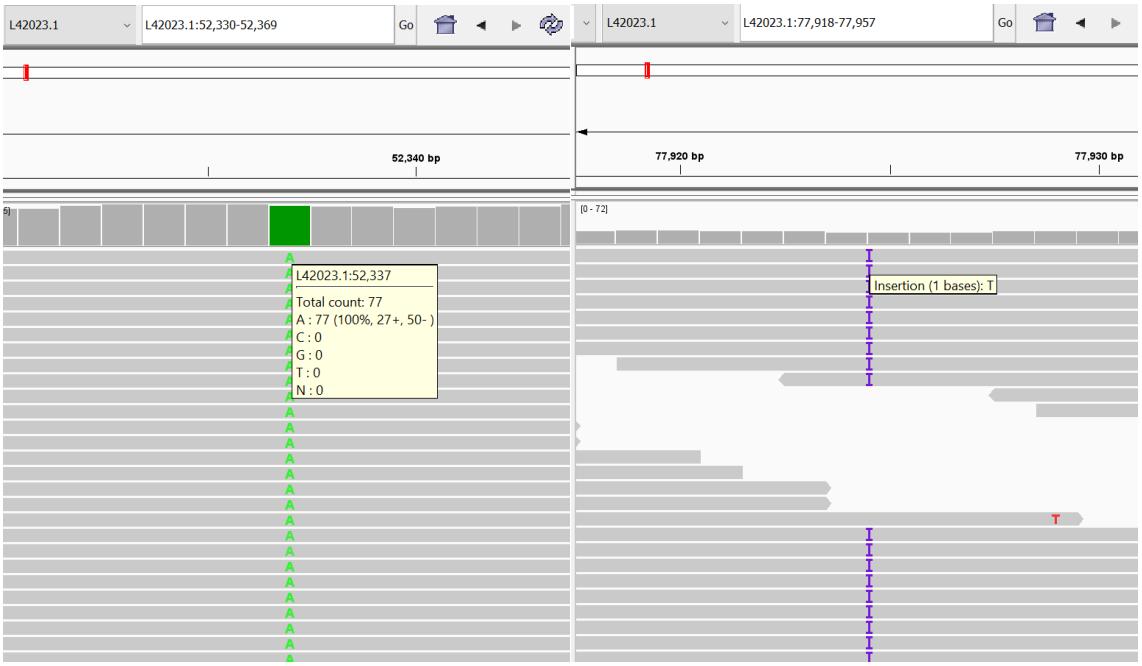
3.1.2 Load your BAM file generated in this module using *File > Load from File >* **HinfKW20_refalign.bam**

3.1.3 Briefly return to Galaxy to view your filtered VCF file. You can pick a site with some interesting variants. In our case we will pick a single nucleotide polymorphism (SNP) at position 52,337.

3.1.4 In the IGV search box, input [redacted] IGV will automatically convert this into a 41bp range centred on your point of interest.

3.1.5 Search for an INDEL site at [redacted] Note the number of reads that cover the site and the reads that support the INDEL call.

How many reads support the presence of an INDEL? How does this compare to our VCF data?




## 3.2.0 Importing your VCF data

Now that you have your aligned reads imported onto the *H. influenzae* genome you can also import your VCF file for comparison. As we will see, however, there are some portions of the BCFtools VCF that IGV is unable to interpret due to the presence of ambiguous nucleotide symbols.

3.2.1 Load your VCF file using *File > Load from File >* **HinfKW20_filtered.vcf**. You will produce an error in IGV during import. The error provided leads us to a possible problem with line 69.

3.2.2 Load your HinfKW20_filtered.vcf in a text editor and search for line 69.



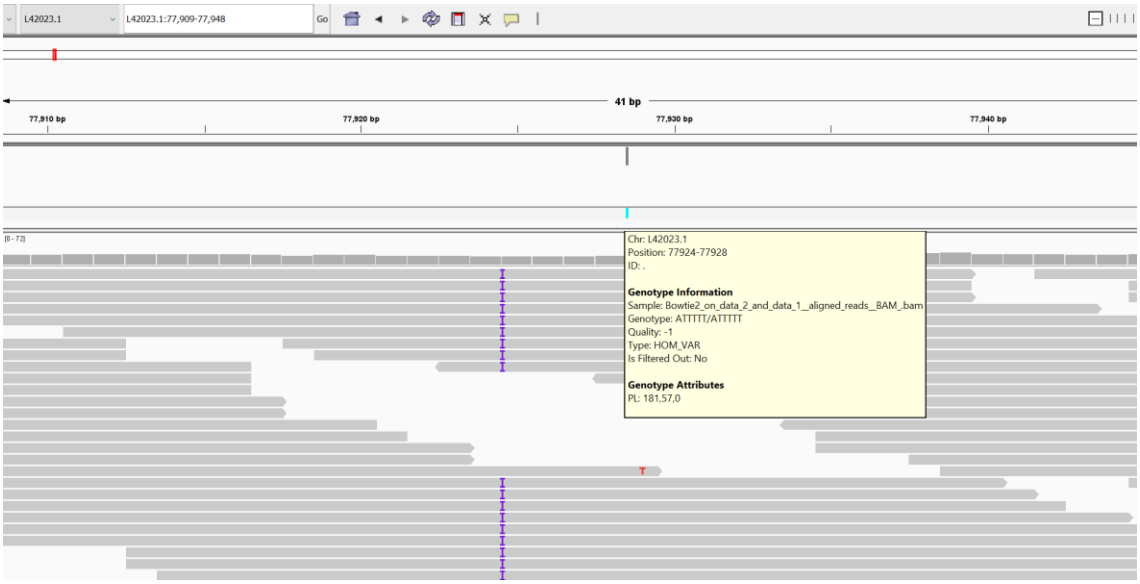
University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

3.2.3

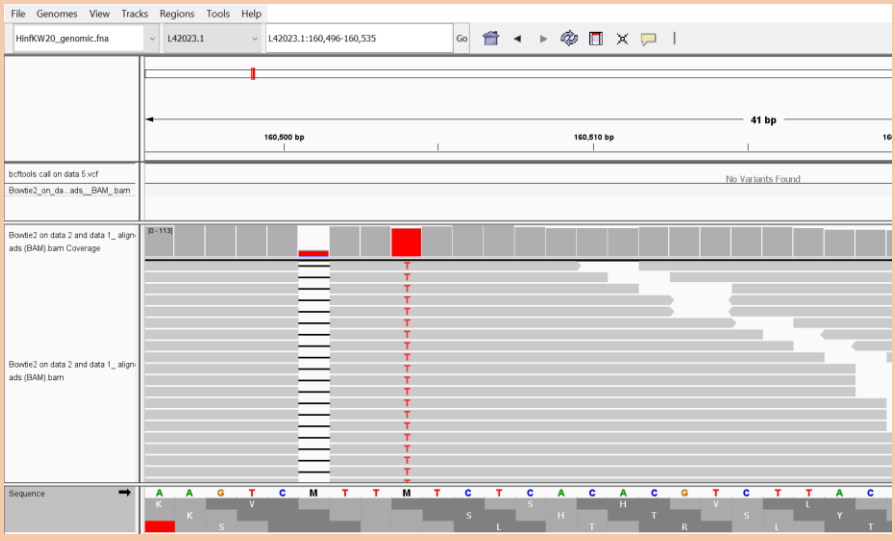
3.2.4

3.2.5 Attempt to load the edited VCF file using *File > Load from File >* **HinfKW20_filtered_edit.vcf**.

3.2.6 Look around the genome with IGV and return to *L42023.1:77,924*. You can now see the VCF entry for this deletion, providing a quick summary about the evidence supporting the call.




**Not all genomes assemblies are perfect!** Recall that our genome assemble for *H. influenzae* originates from the NCBI. Regardless of where we downloaded it, you may find that this file is not necessarily compatible with other programs as we see in the results of our *bowtie2* alignment. Other IUPAC base symbols like K or M, may result in slightly offset or incorrect alignments of reads. The resulting BAM and downstream VCF file created a few entries that IGV could not interpret. What do you think is happening at 160,501 in our reference genome?






University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA




## 3.3.0 Save your session

You can return to this session later to look closer at other variants and examine the genome. What do you think you'll find at position 1,289,372?

3.3.1 Save the session using *File > Save Session >* **HinfKW20_alignment.xml**.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 4.0.0 RNA-Seq analysis with *HISAT2*, *featureCounts*, and *DESeq2*

In a related vein as aligning whole genome sequencing reads to a reference genome is the aligning of RNA-Seq reads to a reference sequence. While similar in idea, we now have two options for alignment: **genome mapping** to a genomic reference or **transcriptome mapping** to a specific transcriptome. In either case we require a reference sequence much like our genomic alignment. Alternatively, you could attempt a *de novo* analysis by assembling transcript reads into contigs and then using this as your reference for counting transcripts afterwards.

For your first RNA-Seq analysis we are going to use a new RNA-Seq dataset that was collected for *H. influenzae* KW20 under two different antibiotic treatments (**beclomethasone** and **prednisolone**). The goal is to identify whether any genes are differentially regulated during these antibiotic treatments. We will be quantifying the expression of all genes in the transcriptome under both treatments and measuring how gene expression differs between these two treatments. However, it is important to note that in the interest of time, we are only using one replicate from each treatment in this portion of the module. In practice, you would have a least three biological replicates under each condition and include a control set of replicates.

## 4.1.0 Load data for the analysis

Before we begin we will need to pull our data from the Shared History page and put it into a new history called **RNA-SeqAnalysis**.

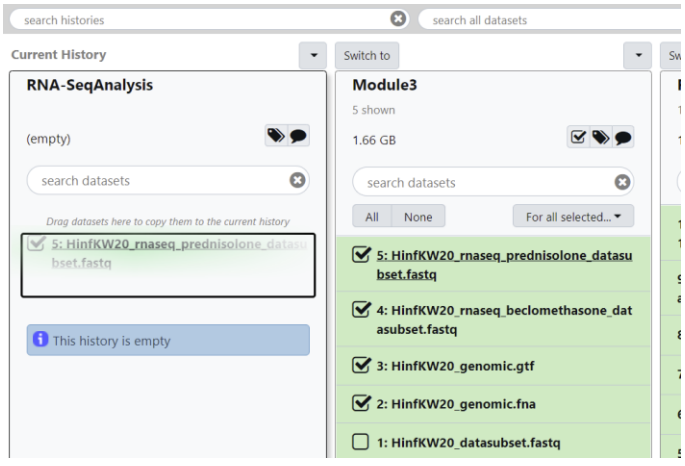
4.1.1 Go to *History pane > Create new history* and name the history **RNA-SeqAnalysis**.

4.1.2 Go to *History pane > Show Histories Side-by-Side*.

4.1.3 Use the **Select histories** button to choose your **RNA-SeqAnalysis** and **Module3Data** histories for viewing.

4.1.4 From the **Module3Data** history, select the *Select Items* icon and select the following history entries:

- [redacted]
- [redacted]
- [redacted]
- [redacted]




4.1.5 Drag and drop the items into your RNA-SeqAnalysis history.

4.1.6 Return to the Galaxy homepage.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

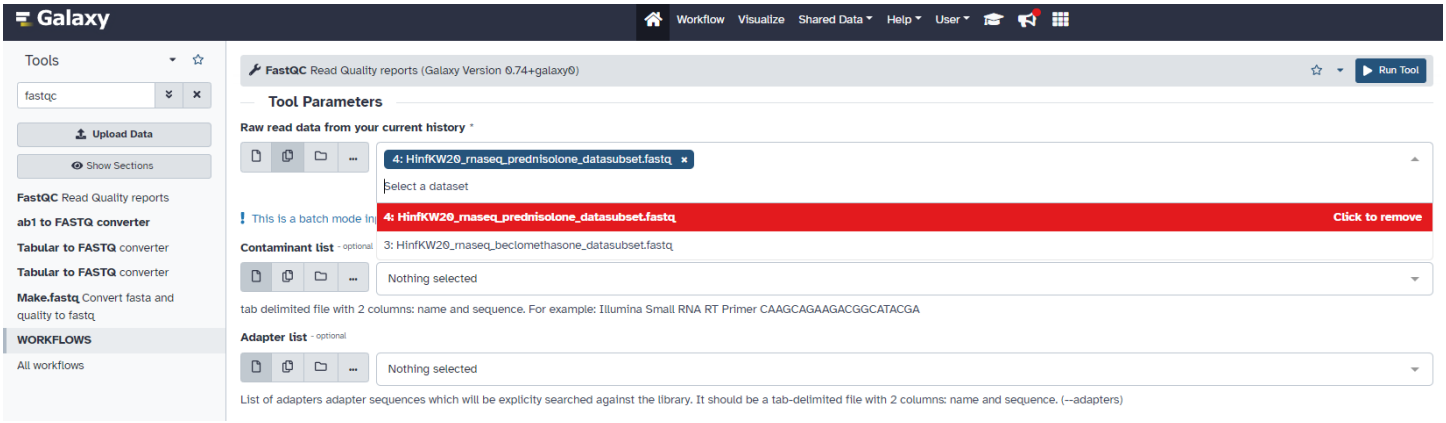
## 4.2.0 Assess the quality of your RNA-Seq data with *FastQC*

Before we go through the effort of analysing these reads, we should assess their quality with FastQC and see how they differ from our random subset of raw genomic data.

4.2.1 Select *tools pane > FastQC Quality Control >* ***FastQC Read Quality reports***

4.2.2 Execute *FastQC* with the following settings:

- Short read data from your current history: **multiple datasets (dual folder icon)**
  - Select both RNA-seq datasets by clicking on them.
- Other options: **default**




## 4.3.0 Review the RNAseq FastQC output

4.3.1 Review the HTML output from both data files by selecting the *View data* icon on the history entry.

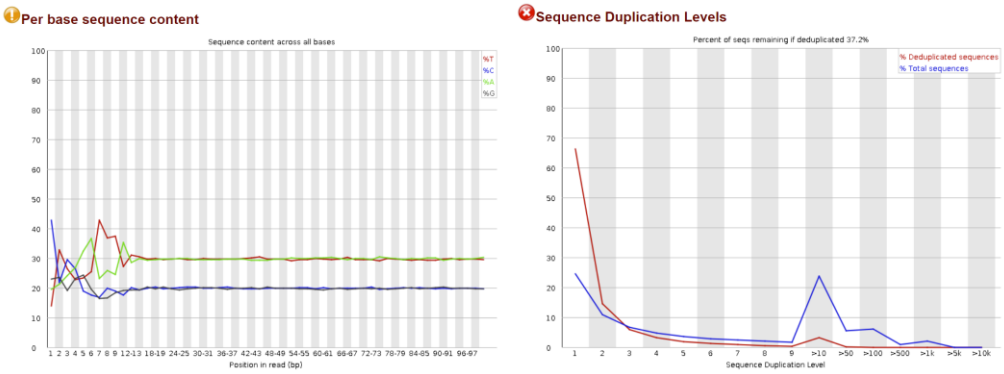
- **5: FastQC on data 2: Webpage**
- **7: FastQC on data 3: Webpage**

You should notice that the overall quality of the sequencing data is good, but that there are a few tests that produce warnings or fail (**Per base sequence content**, **Sequence Duplication Levels**, and **Overrepresented sequences**). When looking at the plots, you will notice that the per base sequence content warning is caused by the same issue that we saw in Module 2 – bias in the base content at the start of reads. As we noted in Module 2, this tends to be indicative of slight enzyme and/or hexamer bias in library preparation, and we may choose to leave this alone.

As for the **Sequence Duplication Levels** and **Overrepresented sequences**, this is a common observation in RNA-Seq data because the expression levels of different genes can vary by *orders of magnitude*, so you would expect to encounter more duplicated reads from those genes. There is considerable debate on the extent to which you should deduplicate RNA-Seq data, with some lines of research suggesting that you shouldn't deduplicate at all. A well-made library should not have much in the way of overduplication of fragments. For simplicity, today we will proceed without deduplicating our dataset, but we will look for cases where there is unequal coverage within genes when we visualize our alignments.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA




## Overrepresented sequences

| Sequence | Count | Percentage | Possible Source |
|---|---|---|---|
| CTTCGATCCTCAAACGGTGGCTTCCACACGATCTCGTTTTGGTTTGACTC | 4501 | 0.22505 | No Hit |
| CTGGATTCGACGGGATTAGCGAAGCCCAAGGTGCACGTCGAGGTGCGGTA | 4172 | 0.2086 | No Hit |
| CTCGTCTTTAATTTCACTTAAGCATGCGGACAGACACGCTAAACTTAAGC | 4089 | 0.20445 | No Hit |
| GTCCGAAATTACTCTACCTTCAGTACTACACGTTTAGTCTCGTCTTTAAT | 3474 | 0.1737 | No Hit |
| CGGGATTAGCGAAGCCCAAGGTGCACGTCGAGGTGCGGTAGGCCTCGTAA | 3151 | 0.15755 | No Hit |
| CAATCGCTGGTTTATTGAAGCCCTTAACCGTATTTATACGACCTAGTGGG | 2520 | 0.126 | No Hit |
| CGCGGAGTCAAACCAAAACGAGATCGTGTGGAAGCCACCGTTTGAGGATC | 2471 | 0.12355000000000001 | No Hit |
| CTCGTTTTGGTTTGACTCCGCGTTATCCCCGTCTTACGAGCGGAAGCTGG | 2016 | 0.1008 | No Hit |

## 4.4.0 Align your RNA-Seq reads with *HISAT2*

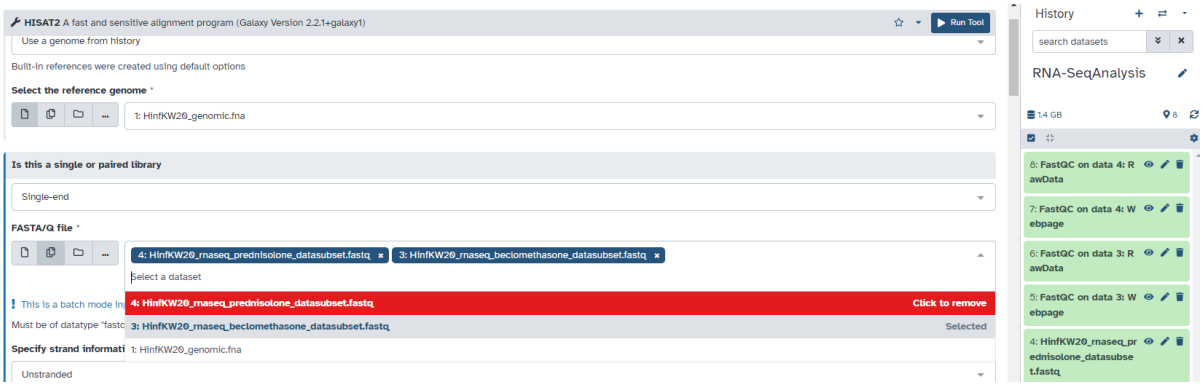
Now that we've looked at the quality of our reads, we can align them to a genome to create a BAM file. This is very much like our previous reference alignment. We'll even use the same reference sequence ***but*** now we will only achieve alignment to the coding regions of the genome (assuming no genomic contamination!).

4.4.1 Select *tools pane > Mapping >*

4.4.2 Scroll down to review the purpose and input/output information for *HISAT2*.

4.4.3 Execute *HISAT2* with the following settings:

- Source for the reference genome: **Use a genome from history**
  - **HinfKW20_genomic.fna**
- Is this a single or paired library: **Single-end**
- 
- Other options: **default**






University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

4.4.4 Note that *HISAT2* will create a BAM file for each RNA-Seq dataset which you will rename using the *History pane > Edit attributes* button (pencil icon).

- After renaming a dataset attribute, be sure to use the *Save* icon!

| History name | New name |
|---|---|
| HISAT2 on dataset 1 and 3: aligned reads (BAM) | HISAT2 on dataset 1 and beclo_R2: aligned reads (BAM) |
| HISAT2 on dataset 1 4: aligned reads (BAM) | HISAT2 on dataset 1 and pred_R3: aligned reads (BAM) |

## 4.5.0 Convert your alignments into count data with *featureCounts*

Now that we've completed a reference alignment of our RNA-Seq reads, we can begin an analysis of those alignments to produce counts. Depending on the algorithm these will analyse the depth of the sequencing reads per segment of genome. Two helpful tools are *featureCounts* and *htseq-count*, both of which are quite efficient for mapping gene counts to a known transcriptome where reads will map uniquely to a single transcript. Other tools such as *StringTie* can produce novel transcripts and annotation for genes that may not exist in a supplied reference file.

Since we are working with a prokaryotic genome, we aren't expecting any alternative splicing or isoforms to deal with. Therefore, we'll be working with the *featureCounts* tool which will produce output compatible for further downstream analysis as well. The read counts produced by this tool are raw, unnormalized read counts.

**What are the normalization units for RNA-Seq counts**? When genome reads are analysed for depth, one should consider the length of genes and the probability of sequencing their entire coding sequence from a library. These values can be influenced by gene length, and the overall read depth of your library. For more information on these units of measure check out this helpful primer: https://www.reneshbedre.com/blog/expression_units.html

4.5.1 [redacted]

4.5.2 Scroll down to review the purpose and input/output information for *featureCounts*.

4.5.3 Execute *featureCounts* with the following settings:

- Alignment file: **Multiple datasets (dual folder icon)**
  - Select both BAM datasets using multi-select (checkbox selection).
- [redacted]
- GFF feature type filter: **gene**
- GFF gene identifier: **gene_id**
- Output format: **Gene-ID "\t" read-count (MultiQC/DESeq2/edgeR/limma-voom compatible)**
- Other options: **default**

4.5.4 Note that *featureCounts* will create two files for each *HISAT2* BAM dataset: a **Counts** and a **Summary** file.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 4.6.0 Review your *featureCounts* output

4.6.1 Select *History pane > featureCounts on data 2 and data 9: Summary > View*.

Here we can see how many reads were assigned to features in a unique manner. You'll find additional information on unassigned reads and why they may be unassigned.

4.6.2 Select *History pane > featureCounts on data 2 and data 9: Counts > View*.

Here we can see a simple tabular format with the GeneID and read counts listed. If we had chosen the *featureCounts* default output, this would have included additional column information based on the GTF file: Chr, start, end, strand, and length. See below for an example contrasting the two outputs.

| Geneid | HISAT2 on data 1 and pred_R3: aligned reads (BAM) |
|---|---|
| Geneid | HISAT2 on data 1 and pred_R3: aligned reads (BAM) |
| HI_0001 | 5475 |
| HI_0002 | 823 |
| HI_0003 | 194 |
| HI_0004 | 151 |
| HI_0005 | 137 |
| HI_0006 | 1318 |
| HI_0007 | 360 |
| HI_0008 | 267 |
| HI_0009 | 74 |
| HI_0010 | 38 |
| HI_0011 | 127 |
| HI_0012 | 80 |
| HI_0013 | 228 |
| HI_0014 | 159 |
| HI_0015 | 544 |
| HI_0016 | 860 |
| HI_0017 | 2700 |
| HI_0018 | 106 |
| HI_0019 | 1047 |
| HI_0020 | 1305 |

| Geneid | Chr | Start | End | Strand | Length | HISAT2 on data 1 and pred_R3 aligned reads (BAM) |
|---|---|---|---|---|---|---|
| Geneid | Chr | Start | End | Strand | Length | HISAT2 on data 1 and pred_R3 aligned reads (BAM) |
| HI_0001 | L42023.1 | 2 | 1021 | + | 1020 | 5475 |
| HI_0002 | L42023.1 | 1190 | 3013 | + | 1824 | 823 |
| HI_0003 | L42023.1 | 3050 | 3838 | - | 789 | 194 |
| HI_0004 | L42023.1 | 3854 | 4318 | - | 465 | 151 |
| HI_0005 | L42023.1 | 4579 | 5391 | - | 813 | 137 |
| HI_0006 | L42023.1 | 5656 | 8748 | + | 3093 | 1318 |
| HI_0007 | L42023.1 | 8750 | 9688 | + | 939 | 360 |
| HI_0008 | L42023.1 | 9681 | 10397 | + | 717 | 267 |
| HI_0009 | L42023.1 | 10467 | 11375 | + | 909 | 74 |
| HI_0010 | L42023.1 | 11414 | 11854 | - | 441 | 38 |
| HI_0011 | L42023.1 | 11857 | 12261 | - | 405 | 127 |
| HI_0012 | L42023.1 | 12367 | 13359 | + | 993 | 80 |
| HI_0013 | L42023.1 | 13423 | 14331 | - | 909 | 228 |
| HI_0014 | L42023.1 | 14328 | 15011 | - | 684 | 159 |
| HI_0015 | L42023.1 | 15013 | 16062 | - | 1050 | 544 |
| HI_0016 | L42023.1 | 16071 | 17867 | - | 1797 | 860 |
| HI_0017 | L42023.1 | 18035 | 18418 | - | 384 | 2700 |
| HI_0018 | L42023.1 | 18676 | 19335 | + | 660 | 106 |
| HI_0019 | L42023.1 | 19405 | 20829 | - | 1425 | 1047 |
| HI_0020 | L42023.1 | 21248 | 22687 | - | 1440 | 1305 |

4.6.3 Rename the history entries for *featureCounts* to the following:

| History name | New name |
|---|---|
| featureCounts on dataset 2 and X: Counts | featureCounts on dataset 2 and beclo_R2: Counts |
| featureCounts on dataset 2 and X: Summary | featureCounts on dataset 2 and beclo_R2: Summary |
| featureCounts on dataset 2 and Y: Counts | featureCounts on dataset 2 and pred_R3: Counts |
| featureCounts on dataset 2 and Y: Summary | featureCounts on dataset 2 and pred_R3: Summary |

* Note that you must match **X** to beclo_R2 data and **Y** to pred_R3 data

## 4.7.0 Add additional *featureCounts* data to your History

Before we can run our next tool, we'll need to include additional replicates for our dataset. Normally you would have included these in our initial analyses, but to save on time I've included them in their final **featureCounts** form. We'll include these in the next step for visualization and analysis purposes.

4.7.1 Select *activity pane > Histories > Shared with me*

4.7.2 Select *Module3_replicates > View*.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

4.7.3 You will see 4 additional sets of *featureCounts* data and summaries that you can import using the *Import this history* button on the upper-left. Import this as **Module3_replicates** and then *Copy History*

4.7.4 Select *activity pane > History Multiview*.

4.7.5 Use the *Select histories* link and add the **Module3_replicates** history.

4.7.6 *Switch to* **RNA-SeqAnalysis.**

4.7.7 From the **Module3_replicates** history drag and drop all 8 history entries to your **RNA-SeqAnalysis** history. (*Select Items > Select All > Drag and drop*)

4.7.8 Return to the Galaxy homepage.

## 4.8.0 Use MultiQC to quickly summarize multiple datasets

When you are working with many conditions and their replicates, as would be the case with RNA-Seq analysis, it is helpful to have a visual summary of all the samples in a single place. *MultiQC* can accomplish that for you by summarizing data/results across your many datasets. In this case, we will summarize our **featureCounts** results to get a sense of how well the reads aligned to the reference genome features we provided.

4.8.1 [redacted]

4.8.2 Scroll down to review the purpose and input/output information for *MultiQC*.

4.8.3 Execute *MultiQC* with the following settings:

- Results > Which tool was used [to] generate logs?: **featureCounts**
- Software name > Output of FeatureCounts:
  - [redacted].
- Report title: **featureCounts summary analysis**
- Other options: **default**

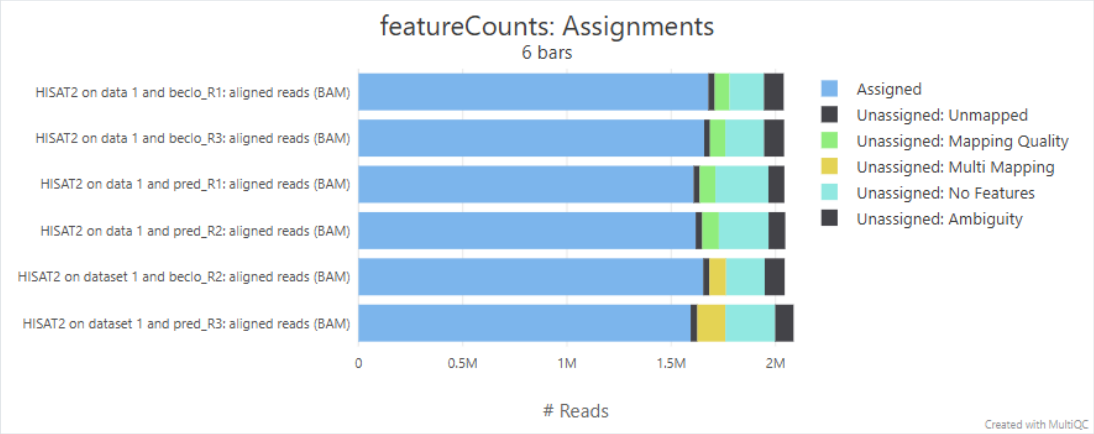



4.8.4 Review the output of *MultiQC* using *History pane > MultiQC on data X, data Y, and others: Webpage > View*. Here you'll be able to see a nice visualization of the assigned and unassigned reads based on category.

4.8.5 You can toggle between percentages and total reads for the figure as well as hover over specific samples to get a per-category breakdown as well.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA




## 4.9.0 Identifying differentially expressed genes with DESeq2

Now that we have multiple replicates we can proceed with a proper analysis of our RNA-Seq data. Normally, you would be able to run DESeq2 without replicates – while not recommended this can be useful for exploratory analysis. The Galaxy interface, however, has removed the option to perform such a run without replicates for computation and technical reasons. We'll see in later modules how to accomplish this from the command line. For now, we'll run with all our replicate data.

**Didn't you say we need normalized data?** *DESeq2* runs the normalization of your read counts ***internally*** using the data that has been provided. Therefore, you can provide the raw read counts generated by *featureCounts*. For more information on this check out the DESeq2 manual here.

4.9.1 Select *tools pane > RNA-seq >* ***DESeq2 Determines differentially expressed features from count tables***

4.9.2 Scroll down to review the purpose and input/output information for *DESeq2*.

4.9.3 Execute *DESeq2* with the following settings:

- 
- 
- 
- 
- Files have header? **Yes**
- Choice of input data: **Count data (eg from HTSeq-count, featureCounts or StringTie)**
- Output options: (to select)
    - Generate plots for visualizing the analysis results
    - Output normalized counts
    - Output VST normalized table



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

4.9.4 The *DESeq2* tool will generate 4 output files for us to review: **DESeq2 result file**, **DESeq2 plots**, a **Normalized counts file**, and a **VST-Normalized counts file**.

## 4.10.0 Review your *DESeq2* output

That takes us to the end of this initial analysis. Let's briefly look at our output and see how the treatments compare.

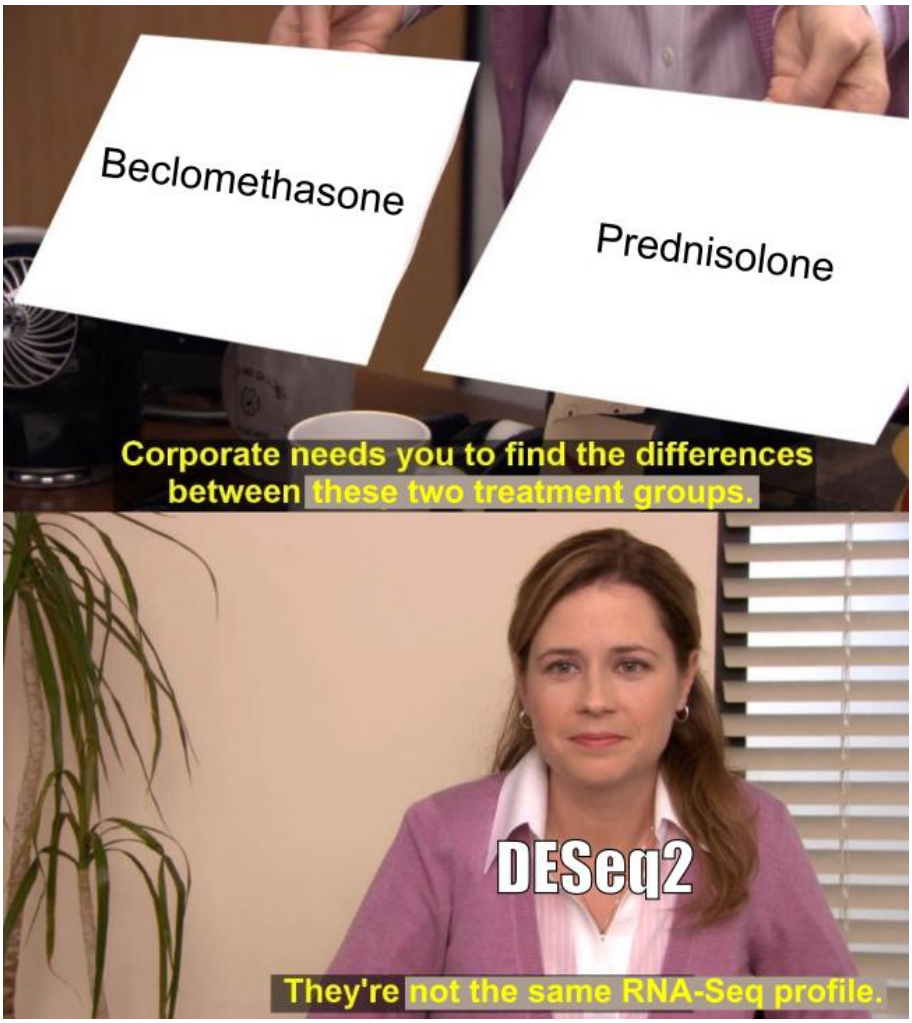
4.10.1 View the plots generated by DESeq2 by selecting *History pane > Display* on the **DESeq2 plots** entry.

4.10.2 View a table of the $\log_2$ fold-change data along with p-values by selecting *History pane > Display* on the **DESeq2 result** entry. You can import these into additional tools for further analysis or use the data to create your own plots! Note that Galaxy no longer provides the headers when viewing this data file but you can see their meaning in the help information found at the bottom of the ***DESeq2*** tool.

| GeneID | Base mean | log2(FC) | StdErr | Wald-Stats | P-value | P-adj |
|---|---|---|---|---|---|---|
| HI_0815 | 7725.19388239346 | -2.674066858433065 | 0.184818219972345 | -11.22220684745811 | 3.172473186066464e-29 | 5.494723548911115e-26 |
| HI_0809 | 17335.4868907276 | -2.697120888639906 | 0.191662696307272 | -10.9417269442822 | 7.27998161729792e-28 | 6.304463566097999e-25 |
| HI_0092 | 536.611411932847 | -2.090268358427851 | 0.205657689112388 | -10.1638970723638 | 2.873557431206654e-24 | 1.659909049628324e-21 |
| HI_0017 | 2610.17417253772 | -1.87142885154697 | 0.18677039129907 | -10.019943945774 | 1.245735776740443e-23 | 5.394035913286604e-21 |
| HI_0053 | 362.4240344449593 | -1.4824096952944 | 0.149486966884383 | -9.91704647215034 | 3.509886339411137e-23 | 1.215826246845450-20 |
| HI_0685 | 683.629761148889 | -2.336488157654456 | 0.237763018648482 | -9.8269620352899 | 8.617927619922216e-23 | 2.487708263819536-20 |
| HI_0181 | 700.4967319887577 | -1.298265869278871 | 0.144783870862957 | -8.96692333254256 | 3.049122073250596-19 | 7.544399186957186-17 |
| HI_0524 | 3046.152905920590 | -0.9609966011967862 | 0.109740596797175 | -8.75697818323325 | 2.005544566891216-18 | 4.342003989917486-16 |
| HI_0813 | 929.2713897086066 | -1.090666233262055 | 0.117330593792476 | -8.52855338797598 | 1.481891056498916-17 | 2.851817619951246-15 |
| HI_0564 | 646.4944291261750 | -1.559462935458026 | 0.183251274510935 | -8.50997947429882 | 1.739776516916526-17 | 3.013292916907386-15 |
| HI_0131 | 6129.536202763956 | -1.0506396049579 | 0.127584537589607 | -8.2348505924552 | 1.797818650372776-16 | 2.830747184041526-14 |
| HI_0812 | 2618.623135906190 | -0.8162176708573630 | 0.100185099385704 | -8.08721460251696 | 6.104465437375146-16 | 8.810778447944796-14 |
| HI_1331 | 294.5554147733670 | 1.410402509244260 | 0.175052280878355 | 8.05703583619318 | 7.816673986492286-16 | 1.041421488046516-13 |
| HI_0629 | 5266.622861053620 | -1.014504916277160 | 0.128456118473765 | -7.897676827938350 | 2.841497296652016-15 | 3.515338684143776-13 |
| HI_0822 | 12494.30166557750 | -1.293856782388790 | 0.164203144010445 | -7.879610285089850 | 3.284037277610746-15 | 3.791968376547876-13 |
| HI_1048 | 660.2640376510650 | 1.217344816499750 | 0.156186089321349 | 7.79419487221487 | 6.482044563554886-15 | 7.016813240048166-13 |
| HI_0091 | 301.6464779048570 | -1.659073357838030 | 0.214106482926913 | -7.748823553609380 | 9.274778728518986-15 | 9.449362798702866-13 |
| HI_0098 | 526.2042462797170 | 1.471713127279520 | 0.193658590269484 | 7.623142413308820 | 2.475734789170336-14 | 2.382207030468346-12 |
| HI_0751 | 2382.385229210620 | -0.8059098709727900 | 0.106346033069918 | -7.569721669293750 | 3.749247226555996-14 | 3.409530629681576-12 |



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA






University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 5.0.0 Review your RNA-Seq analysis in IGV

Now that we've completed our analysis, we can download all of the generated data to our local directories and view them in IGV.

## 5.1.0 Download your dataset from Galaxy

5.1.1 Return to your output list on the *History pane* and select the S*elect Items* icon.

5.1.2 Select the following 7 datasets.

- HinfKW20_genomic.gtf
- HISAT2 on data data 1 and [beclo_R2/pred_R3]: aligned reads (BAM) [2 entries]
- featureCounts on data 2 and [beclo_R2/pred_R3]: Counts [2 entries]
- DESeq2 result file [1 entry]
- DESeq2 plots [1 entry]

5.1.3 Select *History pane > 7 of 27 selected > Auto Build List*.

5.1.4 Name the dataset list **RNA-SeqDownload** and uncheck **Hide original elements** box.

5.1.5 Use the *Create list* button.

5.1.6 Unselect the *Select Items* icon.

5.1.7 Select *History pane > RNA-SeqDownload > Download*.

5.1.8 Save the .zip file to **~/FGDS/Module3/downloads** (~215 Mb).

5.1.9 Decompress the contents to **~/FGDS/Module3**.

5.1.10 Rename your files using the following table

| Download name | New name |
|---|---|
| HISAT2 on data 1 and beclo_R2: aligned reads (BAM).bam | HinfKW20_beclo_R2_HISAT2.bam |
| HISAT2 on data 1 and beclo_R2: aligned reads (BAM).bam.bai | HinfKW20_beclo_R2_HISAT2.bam.bai |
| HISAT2 on data 1 and pred_R3: aligned reads (BAM).bam | HinfKW20_pred_R3_HISAT2.bam |
| HISAT2 on data 1 and pred_R3: aligned reads (BAM).bam.bai | HinfKW20_pred_R3_HISAT2.bam.bai |
| featureCounts on data 12 and beclo_R2: Counts.tabular | HinfKW20_beclo_R2_fCounts.tsv |
| featureCounts on data 12 and pred_R3: Counts.tabular | HinfKW20_pred_R3_fCounts.tsv |
| DESeq2 result file on data 22, data 20, and others.tabular | DESeq2_results.tsv |
| DESeq2 plots on data 22, data 20, and others.pdf | DESeq2_plots.pdf |
| HinfKW20_genomic.gtf.gtf | HinfKW20_genomic.gtf |



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 5.2.0 Loading your *HISAT2* alignments in *IGV*

5.2.1 Open up IGV and select *File > New Session*.

5.2.1 If necessary, load your genome again from Module 1 using *Genomes > Load Genome from File* > **HinfKW20_genomic.fna**.

5.2.2 Load your HISAT2 BAM files using *File > Load from File*. Multi-select your **HinfKW20_*_HISAT2.bam** files and the **HinfKW20_genomic.gtf** file as well.

5.2.3 Expand the **HinfKW20_genomic.gtf** track.

## 5.3.0 Explore your HISAT2 alignments

5.3.1 Use the Search bar to locate the feature [redacted]

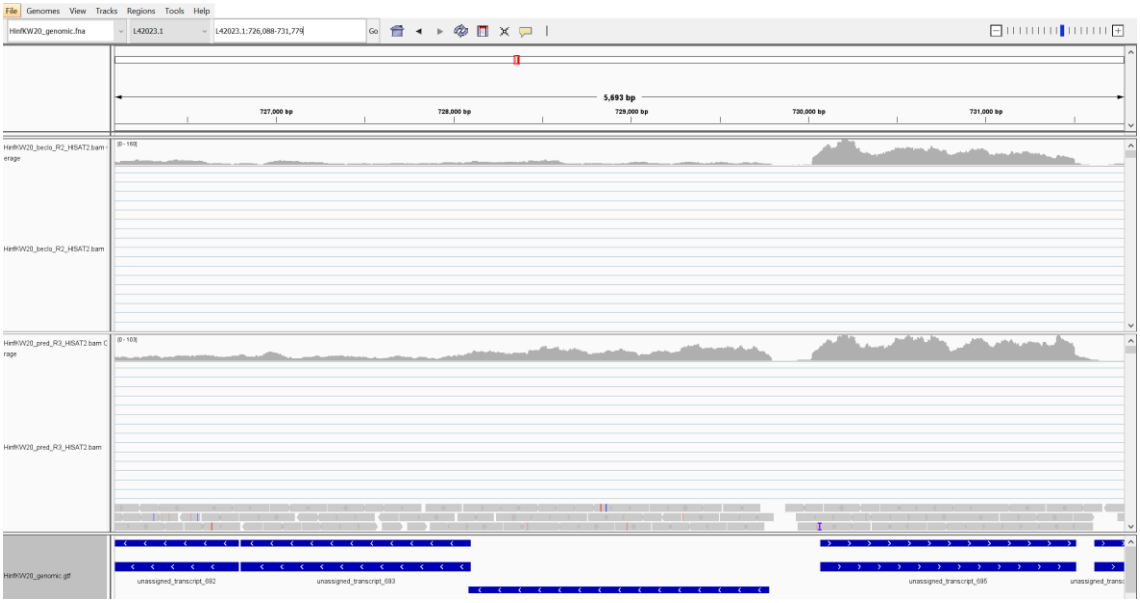
Zoom out to about 50% and notice that this gene appears to have considerably more reads aligned to it in both treatments than surrounding genes, suggesting high expression in both treatments.

5.3.2 Use the Search bar to locate the feature [redacted]

Zoom out to about 50% and notice that HI_1457 appears to be differentially expressed between our two treatments. In HI_1457, the prednisolone treatment appears to produce an upregulation in *H.influenzae* KW20 or beclomethasone may downregulate this gene. A proper untreated control and additional replicates would be used to help examine this effect further.

5.3.3 Use the Search bar to locate the feature [redacted]

Zoom out to about 50% and see another great example of differential expression. What do you think the adjusted p-value on this difference is according to *DESeq2*?




5.3.4 Save your session with *File > Save Session* > **RNASeq.xml**



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 6.0.0 Class summary

That concludes our third lecture and introduction to reference alignment and RNA-Seq analysis on the Galaxy platform. Next module, we will begin our journey into exploring the command-line, its basic commands and useful tools. Altogether we've explored the following in this module:

- Reference sequence alignment using *Bowtie2*.
- Converting BAM files into ***mpileup*** and ***VCF*** formats for analysis.
- Loading BAM and VCF files in *IVG* for visual analysis
- Generating RNA-Seq alignments to a reference genome with *HISAT2*.
- Producing raw read counts with *featureCounts*.
- Summarizing multiple datasets with *MultiQC*.
- Applying differential expression analysis with the *DESeq2* package.
- Reviewing and investigating *HISAT2* RNA-Seq alignments with *IGV*.

## 6.1.0 Post-lecture assessment (9% of final grade)

Soon after this lecture, a homework assignment will be made available on Quercus in the assignment section. It will build on the ideas and/or data generated within this lecture. Each homework assignment will be worth 9% of your final mark. If you have assignment-related questions, please try the following steps in the order presented:

- Check the internet for a solution – read forums and learn to navigate for answers.
- Generate a discussion on Quercus outlining what you've tried so far and see if other students can contribute to a solution.
- Contact course teaching assistants or the instructor.

## 6.2.0 Suggested class preparation for Module 4

Next week we will begin exploring the command-line interface and learning navigate our way around it. To prepare for this, we suggest the following Coursera Modules:

- Command Line Tools for Genomic Data Science, ***Lilana Florea, PhD***:
    - Module 1: Basic Unix Commands (1hr, 40mins)

## 6.3.0 Acknowledgements

This course was originally conceived and produced by Dr. David Guttman and Dr. Marcus Dillon.

- **Revision 1.0.0**: edited and prepared by David Guttman, Ph.D. and Marcus Dillon, Ph.D.
- **Revision 2.0.0**: edited and prepared for *CSB1021HF LEC0131, 10-2021* by Calvin Mok, Ph.D., Education and Outreach, CAGEF.
- **Revision 2.0.1**: edited and prepared for *CSB1021HF LEC0131, 10-2023* by Calvin Mok, Ph.D., Education and Outreach, CAGEF.
- **Revision 2.1.0**: edited and prepared for *CSB1021HF LEC0131, 10-2024* by Calvin Mok, Ph.D., Education and Outreach, CAGEF.
- **Revision 2.1.1**: edited and prepared for *CSB1021HF LEC0131, 10-2026* by Calvin Mok, Ph.D., Education and Outreach, CAGEF.



University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA

## 6.4.0 References


- Danacek P., et al. (GigaScience 2021). *Twelve years of SAMtools and BCFtools.* DOI: 10.1093/gigascience/giab008

- Ewels P., et al. (Bioinformatics 2016*). MultiQC: Summarize analysis results for multiple tools and samples in a single report*. DOI: 10.1093/bioinformatics/btw354

- Kim D., et al. (Nat Biotechnology 2019*). Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype.* DOI: 10.1038/s41587-019-0201-4

- Langmead B., Salzberg S.L. (Nat Methods 2012). *Fast gapped-read alignment with Bowtie 2.* DOI: 10.1038/nmeth.1923

- Liao Y., Smyth G.K., Shi W. (Bioinformatics 2014). *featureCounts: an efficient general purpose program for assigning sequence reads to genomic features.* DOI: 10.1093/bioinformatics/btt656

- Love M.I., Huber W., Anders S. (Genome Biology 2014). *Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2*. DOI: 10.1186/s13059-014-0550-8







University of Toronto Centre for the Analysis of Genome Evolution & Function (cagef.utoronto.ca) Licensed under CC BY-NC-SA